Benjamin L. Davis, Ph.D.

bendavis007 | bendavis007.github.io | BenDavis@MissouriState.edu | (417) 836-3324

Summary

My research focuses on using recent developments in artificial intelligence (AI) and machine learning (ML) to solve the most difficult and impactful problems in astrophysics. Particular interests include:

- Symbolic Regression – used to find new (higher dimensional) scaling relations and provide new insights into the astrophysical processes behind these relations.
- Causal Discovery – enables researchers to look beyond correlation and uncover the fundamental mechanisms driving complex systems.

Education

2015 Ph.D. (Space & Planetary Sciences) at the **University of Arkansas**
Thesis: *Logarithmic Spiral Arm Pitch Angle of Spiral Galaxies: Measurement and Relationship to Galactic Structure and Nuclear Supermassive Black Hole Mass*

2008 B.S. (Physics), B.S. (Mathematics), & Minor (Music) at **Pittsburg State University**

Work Experience

Missouri State University Aug 2026 –
Physics, Astronomy and Materials Science Department – Tenure-track, Assistant Professor of Astrophysics

New York University Abu Dhabi Sep 2020 – Aug 2026
Center for Astrophysics and Space Science – Fellow, Research Associate

Swinburne University of Technology Aug 2016 – Jul 2020
Centre for Astrophysics & Supercomputing – Postdoctoral Fellow

Arkansas Tech University Aug 2015 – May 2016
Department of Physical Sciences – Visiting Assistant Professor of Physics

University of Arkansas Aug 2015 – Dec 2015
Department of Physics – Visiting Assistant Professor of Physics

Publications

37 peer-reviewed publications, 2,767 citations, h-index = 22, and i10-index = 32.
ORCID | Google Scholar | NASA ADS | Scopus | Web of Science | ResearchGate
★ Media Highlight – *NASA Telescopes Capture Stellar Delivery Service for Black Hole*

Technical Proficiencies

Programming Languages & Software

`C++` • `Fortran` • `HTML` • `IDL` • LaTeX • `Mathematica` • `MATLAB` • `Python` • `R` • `Supermongo` • `Adobe Creative Cloud` • `DS9` • `Gimp` • `IRAF` • `Microsoft Office Suite` • `PC/Mac/Unix/Linux OS` • `Git`

For my unabridged curriculum vitæ, visit https://bendavis007.github.io/PDFs/CV.pdf.

Last updated: October 12, 2026; the current version is maintained at https://bendavis007.github.io/PDFs/Resume.pdf.